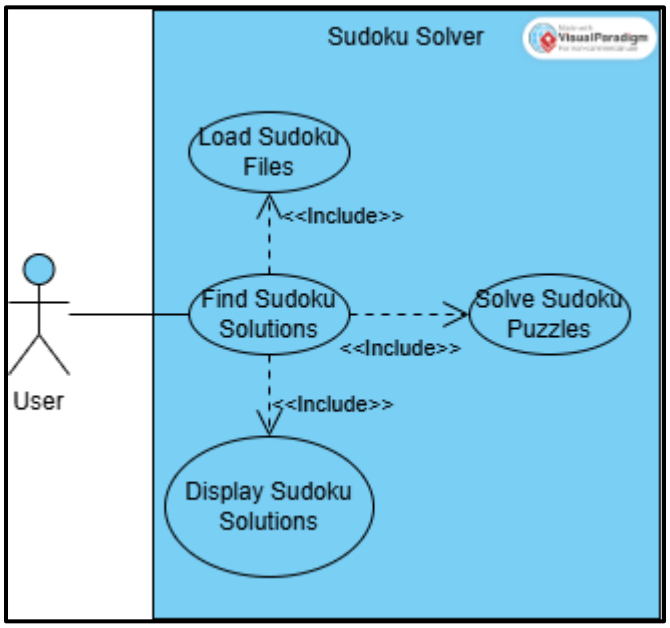
Assignment #4 Use Cases



Find Sudoku Solutions

Brief Description

This use case allows a user to find all solutions to a group of sudoku puzzles.

Flowchart

1. User starts program from console line with arguments that are file paths
2. Program loads in sudoku files from the file paths
3. Program solves the puzzles found in those sudoku files
4. Program displays solutions found in puzzles given by files
5. Program terminates

    1a. If user starts program without arguments, then program informs user of usage before terminating

Detailed Description

User starts the program and provides arguments that are paths to valid sudoku files. If no arguments are provided, then the program will inform the user of its usage before terminating. With files paths, the program will then load those files into the program as sudoku puzzles. If the sudoku files are invalid for any reason, then the program will inform the user of that issue before continuing to the next file provided. Once a file is loaded in, the system then attempts to solve the

puzzle. The program will produce a list of all solutions. After getting this list, the program then will display what it found. It will first display the name of the file it got the puzzle from before printing the original board. From this point, the program will change depending on whether it found any solutions or not. If it did find some number of solutions, then it would print each one with a label based on the order that the solutions were found. If no solutions were found, then the program will print a message informing the user that no solutions were found. After all files are processed, the program will terminate.